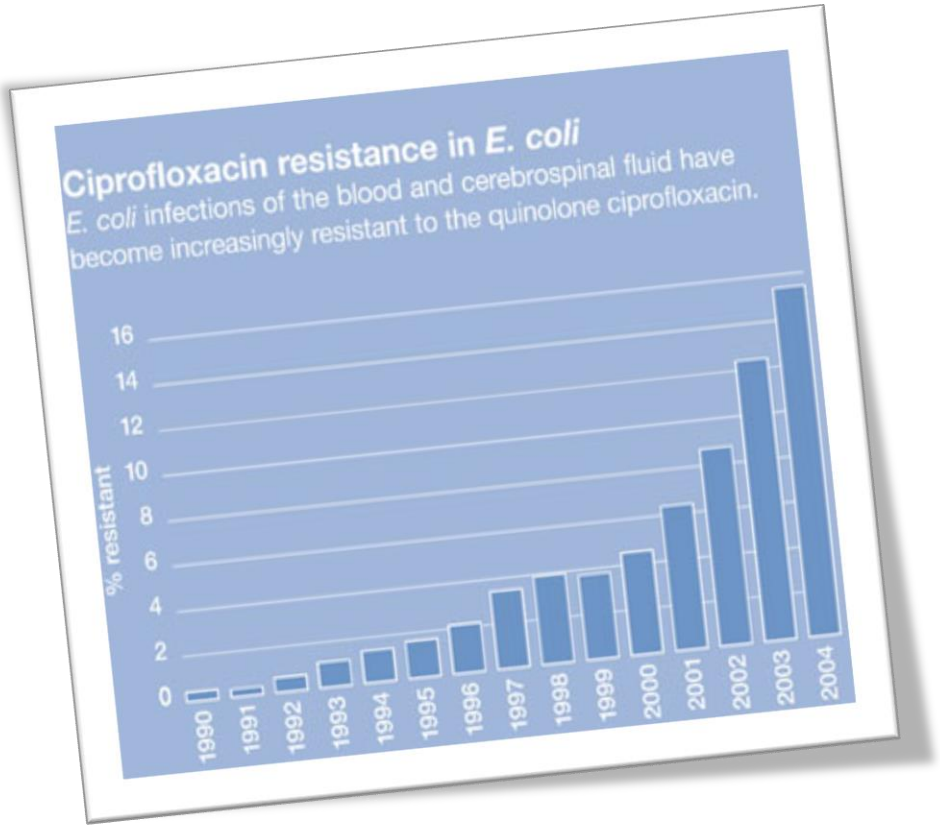


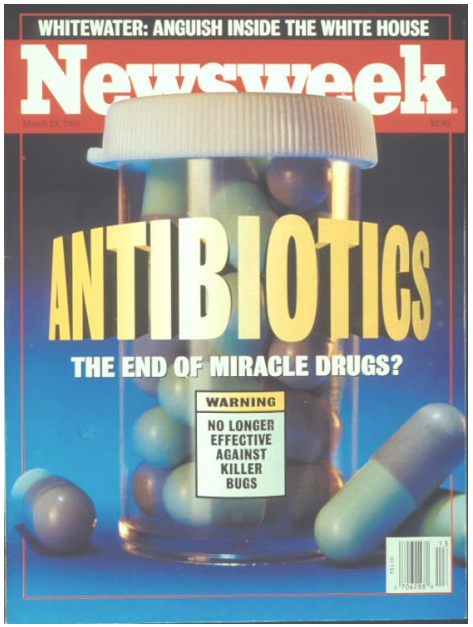
Ontology-mediated Data Integration for real-time Antibiotics Resistance Surveillance

SWAT4LS 2014, Berlin

Dr. Daniel Schober www.ipb-halle.de

Projects Clinical Context





- Steep increase in bacterial antibiotics resistance
- Nasty nosocomial hospital infections

debugIT

A need for ‚IT-biotics‘

- DebugIT
  - Detecting and Eliminating Bacteria UsinG IT
  - ‘ITbiotics’ approach
    - Enable cross-country resistance-data comparison
  - Using ‘semantic data’ to exploit distributed clinical data
    - Apply data in resistance monitoring
- Why are existing surveillance networks insufficient ?

Reviewing existing surveillance efforts

- Paul Ehrlich Gesellschaft (PEG) surveillance
  - Resistance data from <30 German clinics (not covering EU)
  - 3 year interval (low time resolution)
- European Antimicrobial Resistance Surveillance (EARS-NET)
  - European resistance comparison
  - 1 year interval
  - Limited to specific clinical isolates, i.e. blood culture samples
- German Network for Antimicrobial Resistance Surveillance (GENARS)
  - Only German sites
- German Krankenhaus Infektions Surveillance-System (KISS)
  - Only looks at intensive care units
  - Few pathogens
- SARI system
  - Only German sites
  - Low time resolution

Drawbacks →Requirements for DebugIT Surveillance

Limited access to local data-sets

→ Parallel integrated access to EU-wide hospital data

Limited timely availability & coarse time resolution

→ Real-time data access

Limited automatisation

→ Automatic real-time feed of hospital data into vCDR

Limited data coherence, dissemination & re-use

→ Ontologies formalize exchange syntax & domain semantics

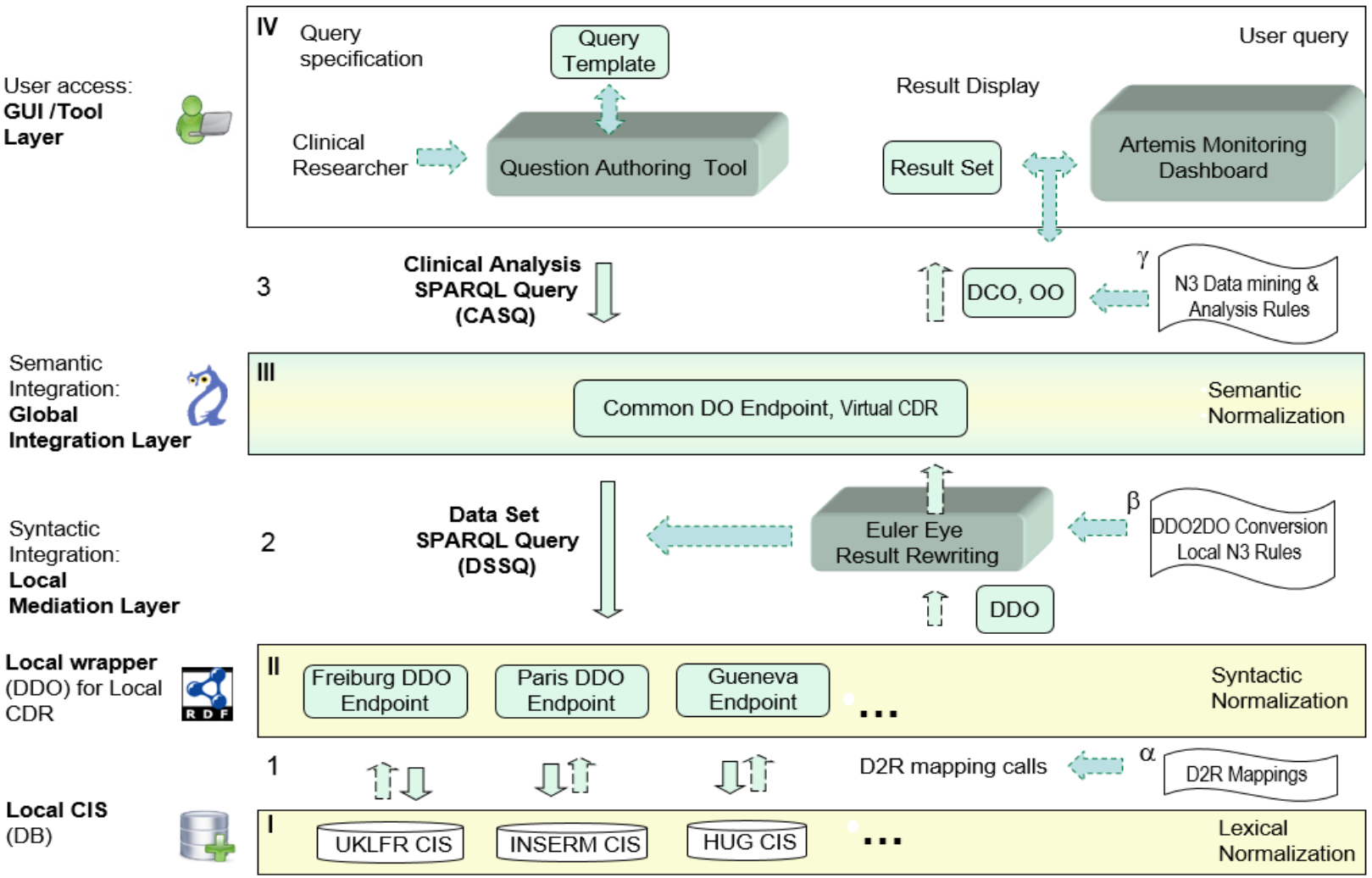
Limited data coverage & granularity

→ Granular domain coverage through expressive ontologies

Methods

- Real-time Data Federation
  - Lexical normalization
  - ETL Process
  - D2R conversion
- Ontologies
  - Integrating geographically distributed, multilingual syntactic & semantically heterogeneous data
    - DDO (Data Definition Ontologies) in N3
    - DCO (DebugIT Core Ontology) in OWL-DL, SKOS
    - Terminological mapping rules, N3
- Reasoning
  - Tableau based DL reasoning (Hermit), T-Box
  - N3 rule-based coherent logics reasoning (Euler-EYE), A-Box
- SPARQL querying
  - On DDO layer for multiple clinical endpoints
  - On DCO layer for vCDR endpoint

DebugIT SIP Architecture



Ontology Layers in WP1a

7 Data Definition Ontologie (DDO) **average 40 Entities**

- **Mediation layer** closing the 'formality'-gap
- Describing **site-specific local** CIS data models in RDF
- For SPARQL data access to local hospital data (DSSQ)

13 Operational ontologies (OO, e.g CAO) **average 35 Entities**

- **Mediation & Syntaxtic Integration layer**
- Implementation, module crosstalk, data mining, evidences, maths, units, …
- OWL-Full →Coherent Logic reasoning (e.g. rule-based)

1 DebugIT Core Ontology (DCO) **~ 1720 Entities**

- **Semantic Integration layer**, mapped to DDOs & external Terminologies
- Rooted in Biotop upper level ontology
- **Global,** clinical domain of infectious diseases
- OWL-DL →DL & Coherent Logic reasoning
- For SPARQL data access on vCDR level (CASQ)

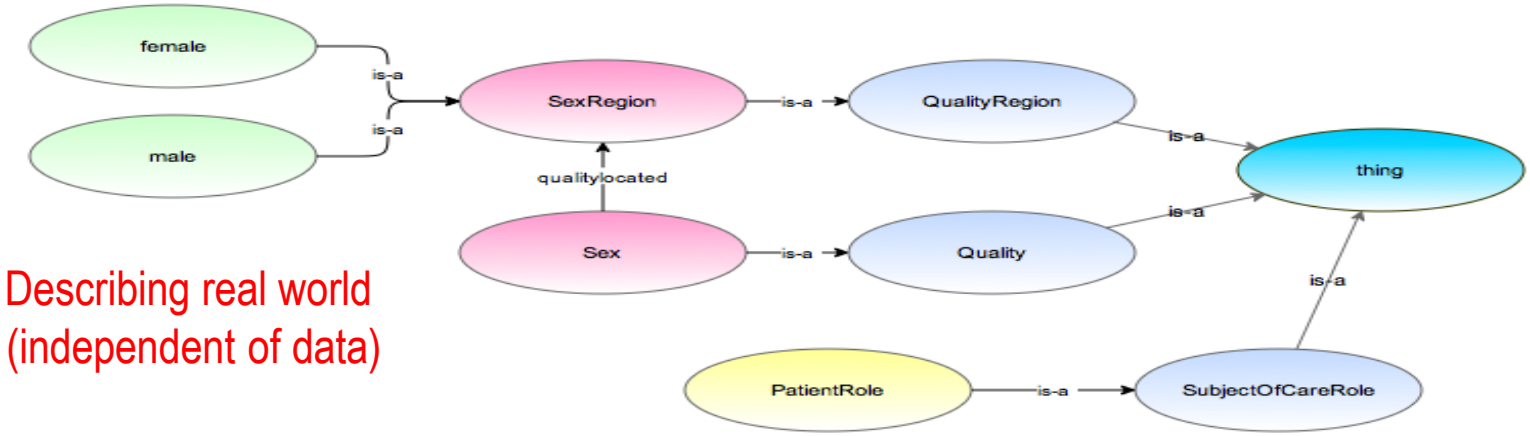


‚Female patient‘ in different ontology layers

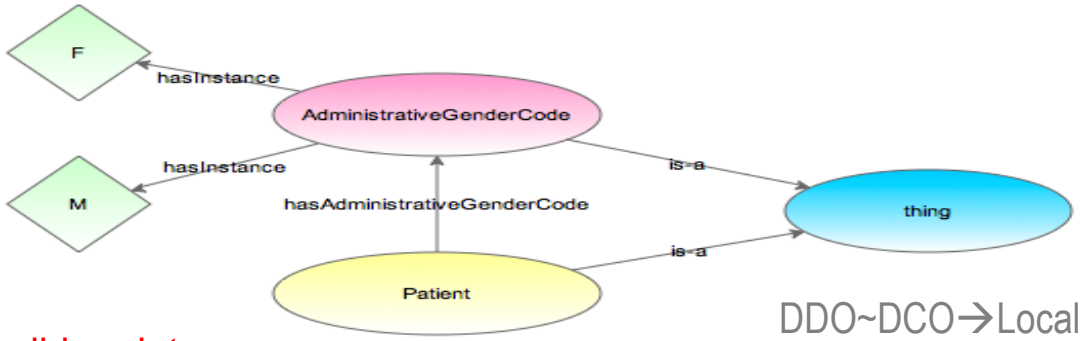
DCO

DDO

E/R



Describing real world (independent of data)



Describing data

DDO~DCO→Local to Global rules to later create SPARQL CONSTRUCT/WHERE clauses

Patient	
ID	FKREFADMINISTRATIVEGENDERCODE
23	F
45	M
58	M

DL Reasoning for DCO maintenance (Infering BloodSample is a BodyLiquidSample)

Stated Facts

BodyLiquidSample =

Sample
and derivesFrom some BodyLiquid

BloodSample =

Sample
and derivesFrom some Blood

BodyLiquid =

- Blood
- BloodSerum
- CerebrospinalFluid
- Sputum
- Urine

Asserted Hierarchy (flat list)

- Sample
  - BloodSample
  - BloodSerumSample
  - BodyLiquidSample
  - DerivedSample
  - FecesSample
  - OrganSample
  - SputumSample
  - SwabSample
  - TissueSample
  - UrineSample

Logics Reasoner



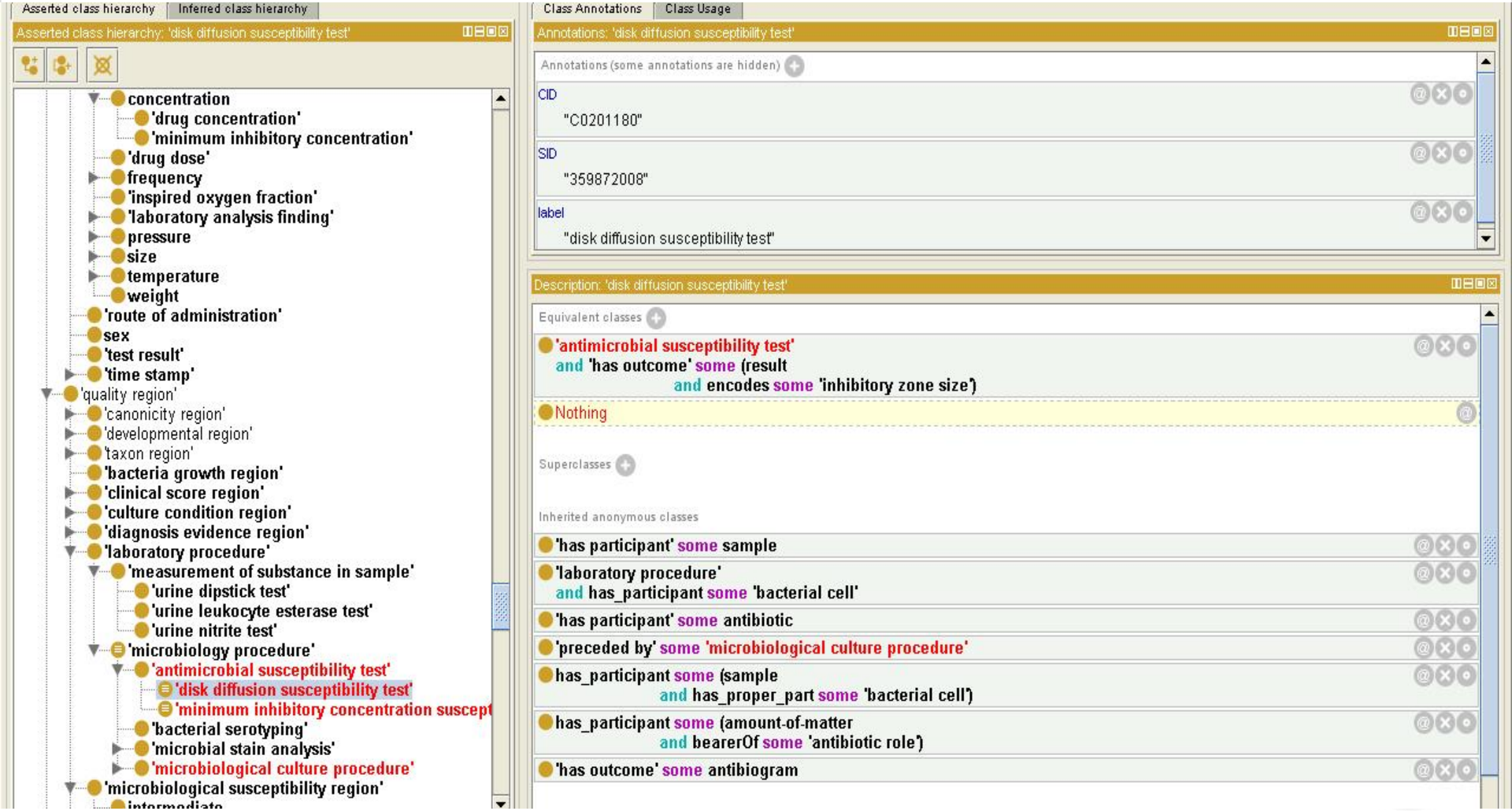
Inferred Hierarchy (more structure)

- Sample
  - BodyLiquidSample
    - BloodSample
    - BloodSerumSample
    - SputumSample
    - UrineSample
  - DerivedSample
  - FecesSample
  - OrganSample
  - SwabSample
  - TissueSample

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Artificial Intelligence helps DCO modeling

- DL semantics used to spot modelling errors
  ‘antimicrobial susceptibility test’ shown to be inconsistent by reasoner



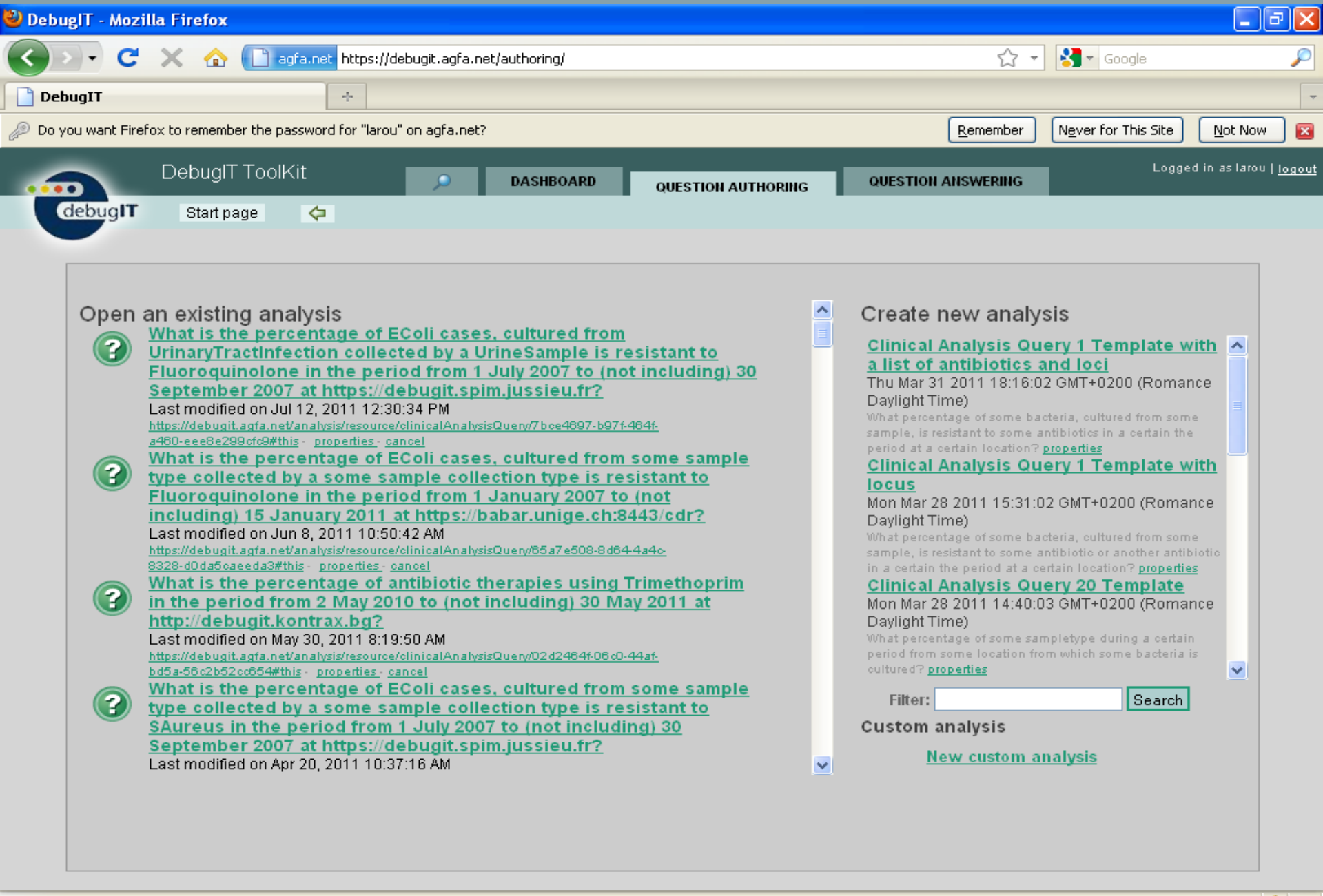
Example Inference (N3 rule)

```
{   ?episodeOfCare a dco:EpisodeOfCare.
    ?bacterialAntibiogramAnalysis
        a dco:BacterialAntibiogramAnalysis;
        biotop:processualPartOf ?episodeOfCare;
        biotop:hasParticipant [biotop:encodes [a ?Antibiotic]];
        dco:hasResultDateTime ?abgResultTime.
    ?abTherapy
        a dco:AntibioticTherapy;
        biotop:processualPartOf ?episodeOfCare;
        biotop:hasParticipant [a ?Antibiotic];
        dco:hasStartDateTime ?therapyStart.
    (?therapyStart ?abgResultTime) math:difference ?dif.
    ?dif math:greaterThan "P0D"^^xsd:duration; math:lessThan "P2D"^^xsd:duration}
=>
{   ?abTherapy
        a dco:AntibioticTherapy;
        event:basedOn ?bacterialAntibiogramAnalysis.
        ?bacterialAntibiogramAnalysis a dco:BacterialAntibiogramAnalysis}.
```

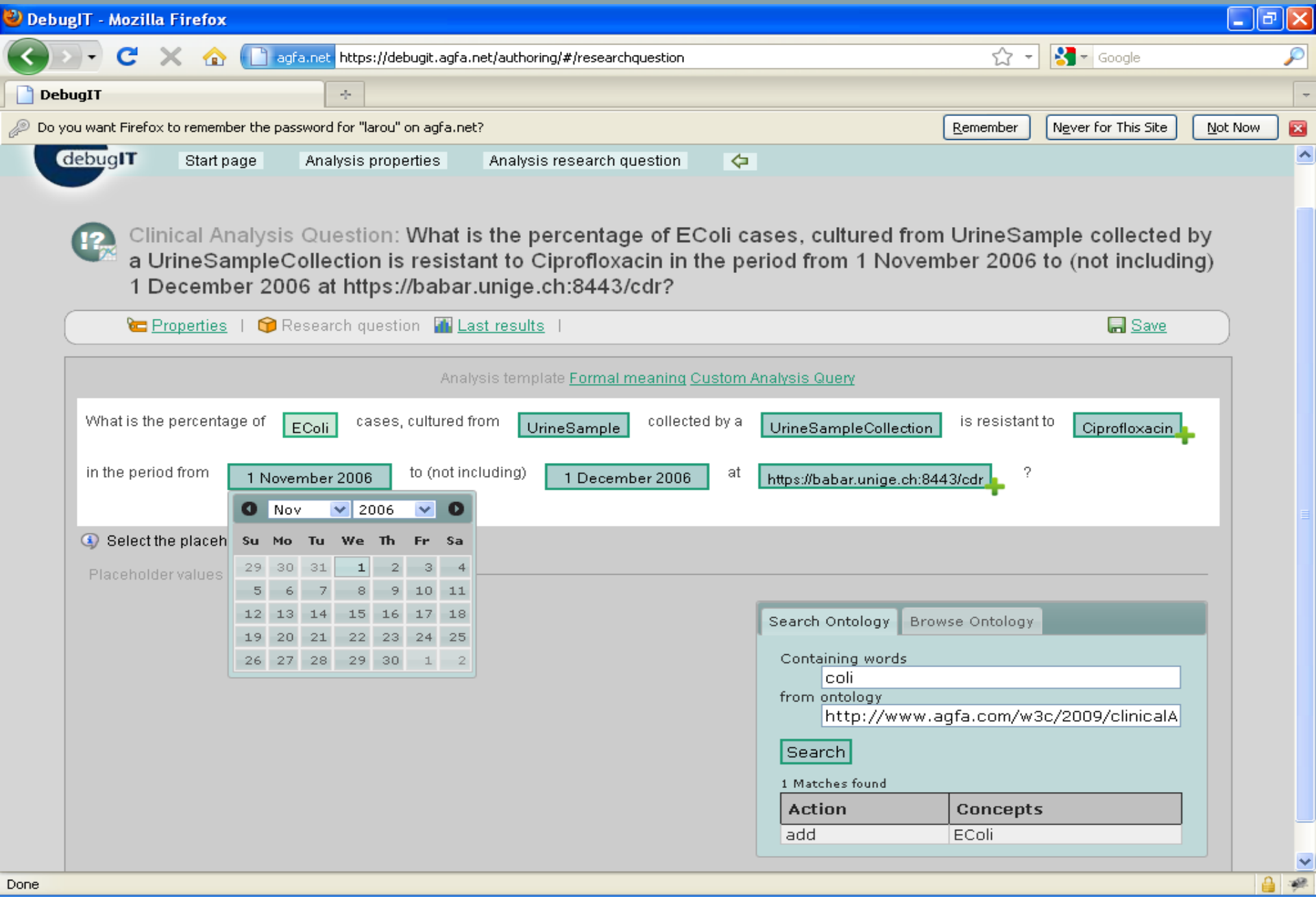
Check if AntibioticTherapy is Antibiogramm based, i.e. started within 2 days after an antibiogram result for same EpisodeOfCare, Bacterium & Antibiotic

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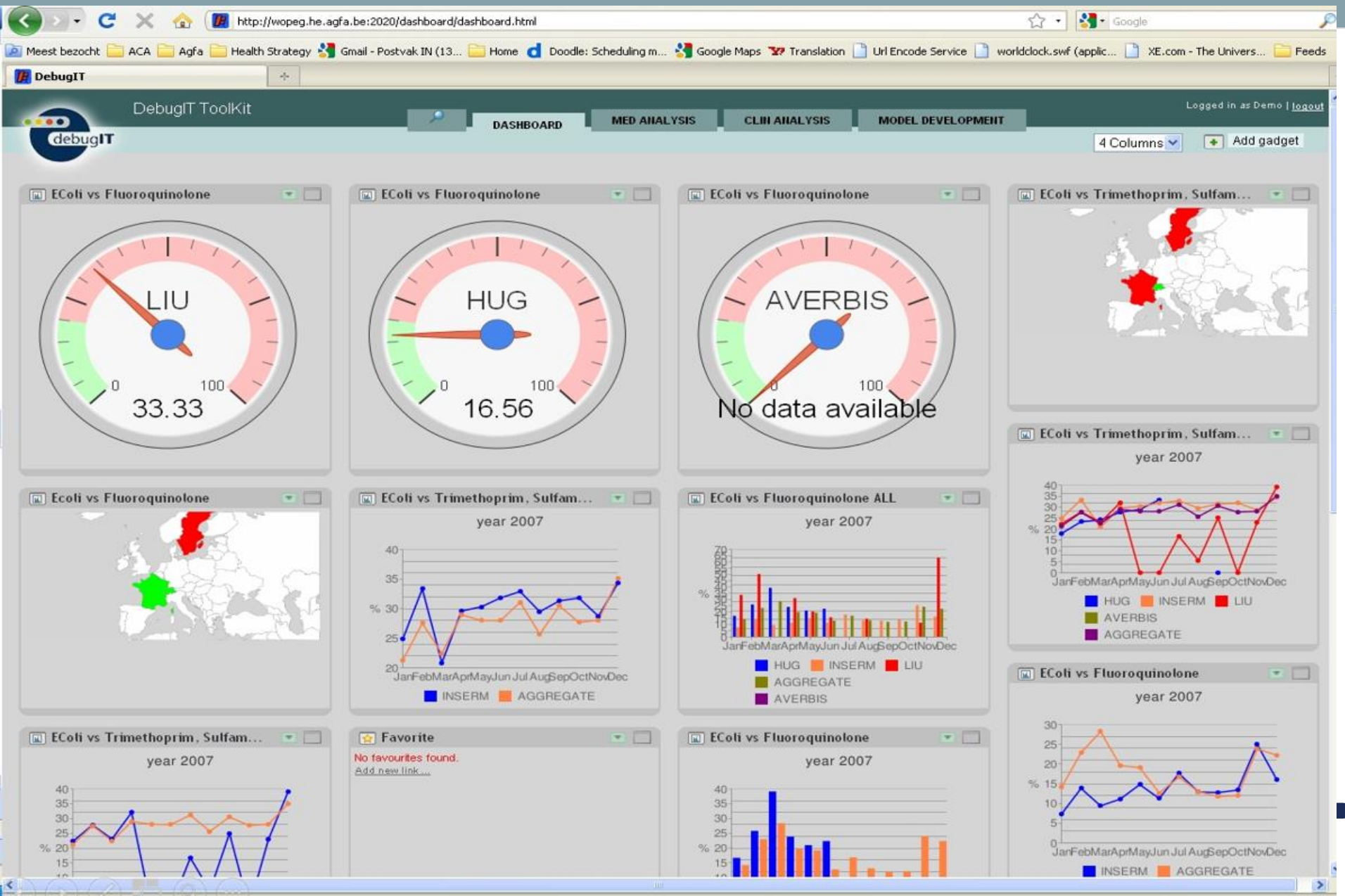
Preconfigured Query Library



Query Template Instantiation



Configurable Resistance Monitoring Dashboard



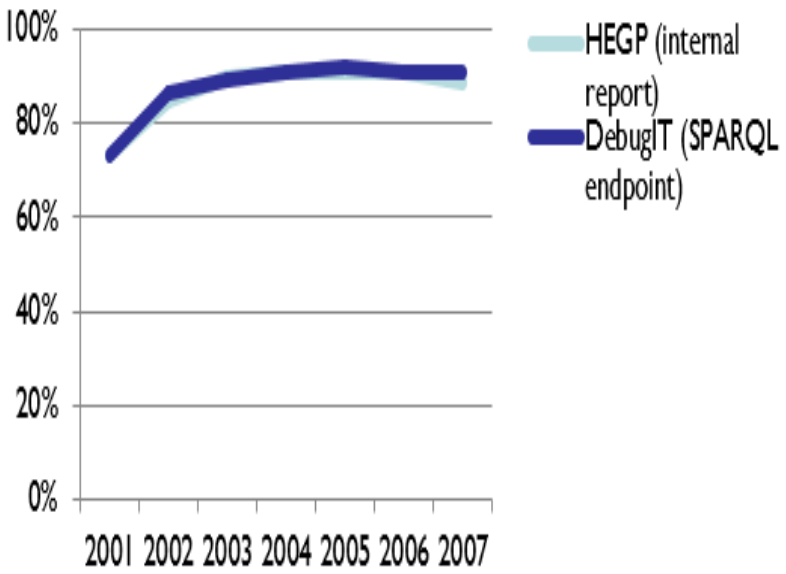
Result Evaluation

- DebugIT vs. HEGP resistance trends for 2001-2007

E. Coli vs CEFIXIME : sensitive @ hegp vs debugit

	2001	2002	2003	2004	2005	2006	2007
HEGP (internal report)	73%	85%	90%	91%	91%	91%	89%
DebugIT (SPARQL endpoint)	73%	86%	89%	91%	92%	91%	91%

total	1244	2404	2853	2780	2838	2850	2727
ddo:Sensitive	911	2074	2553	2525	2607	2596	2479



- Good alignment

  between DebugIT Results & local check by Microbiologists

Conclusion I

- Archived federation of heterogeneous local data into coherent formalized ‘virtual CDR’
- Created semantically & geographically interoperable resistance data
  - Integration across 6 languages over 7 EU hospitals
- Feasibility of semantic integration approach
  - Open & scalable
  - Real-time access
  - Deep-Annotation rather than solely shallow (schema based) annotation

Conclusion II

- Stepwise semantic formalization
  - Bi-layered hybrid formalization approach
    - Bridge semantic gap between local DDO RDF data & global formal DCO OWL integration layer
    - Rule reasoning for layer-binding (DDO to DCO mapping)
    - DL-reasoning for DCO engineering & quality assurance
- Good end user compliance
  - Complexity shielded from user
  - Good tool usability measured via a short questionnaire
    - 10 clinicians, using 5-point Likert scale

Resources & Acknowledgements

DebugIT Resources

- http://www.debugit.eu

Acknowledgements

- Remy Choquet, Hans Cools, Kristof Depraetere, Christian Lovis, Douglas Teodoro, Emilie Paasche, Giovanni Mels, Frank Enders, Philipp Daumke, Daniel Karlsson, Steffen Neumann, Stefan Schulz, Ilinca Tudose, Maren Kechel, Jessica Bullenkamp, Patrick Ruch, Jos De Roo, Marie-Christine Jaulent, Dirk Coalert, Martin Boeker

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Reasoner

- **Formal logical inferring**
  - Adhering coherent logic (close to first order logic)
- **EYE: Euler Yap Engine**
  - Open source: http://eulersharp.sourceforge.net/
  - Inference engine supporting logic based proofs
  - Backward-chaining reasoner enhanced with Euler path detection
  - Implementation in Java/Prolog (logic programming language)
  - **60 million lips** (logical inferences / second) on an Intel Core Duo 2.2 GHz

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